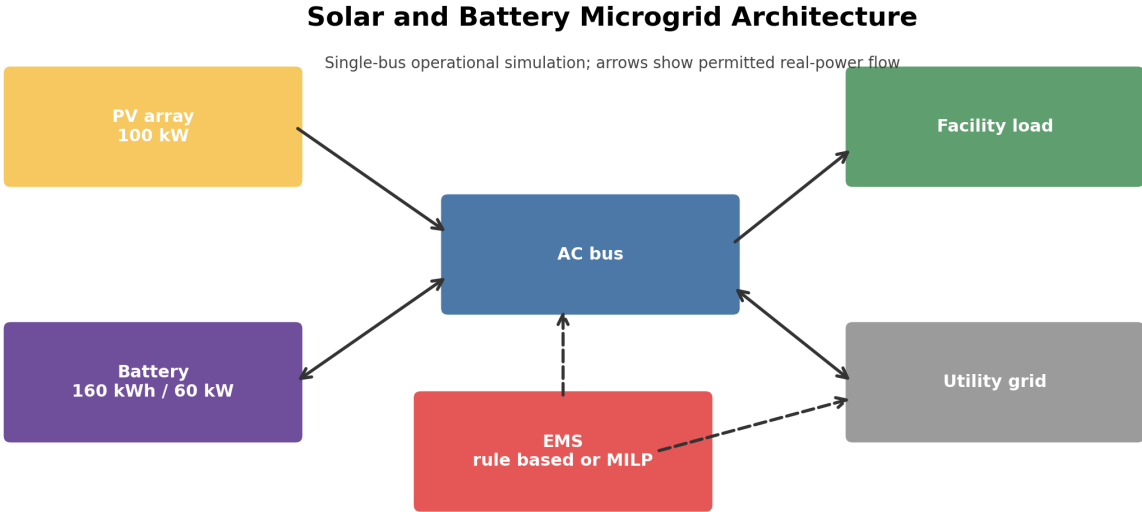


Solar and Battery Microgrid Energy Management Simulator

Technical report | Giorgi Gogitidze | September 2026



This report documents a reproducible software simulation, not physical hardware testing. It compares grid-only, solar-only, rule-based storage, and mixed-integer optimized storage over a synthetic seven-day dataset.

Model and formulation

A single AC bus connects facility load, PV, battery, and utility. The hourly balance is $PV_{used} + import + discharge + unserved = load + charge + export$. Battery energy follows $SOC[t+1] = SOC[t] + \eta_c \cdot charge_{dt} - discharge_{dt} / \eta_d$. A binary mode prevents simultaneous charge and discharge.

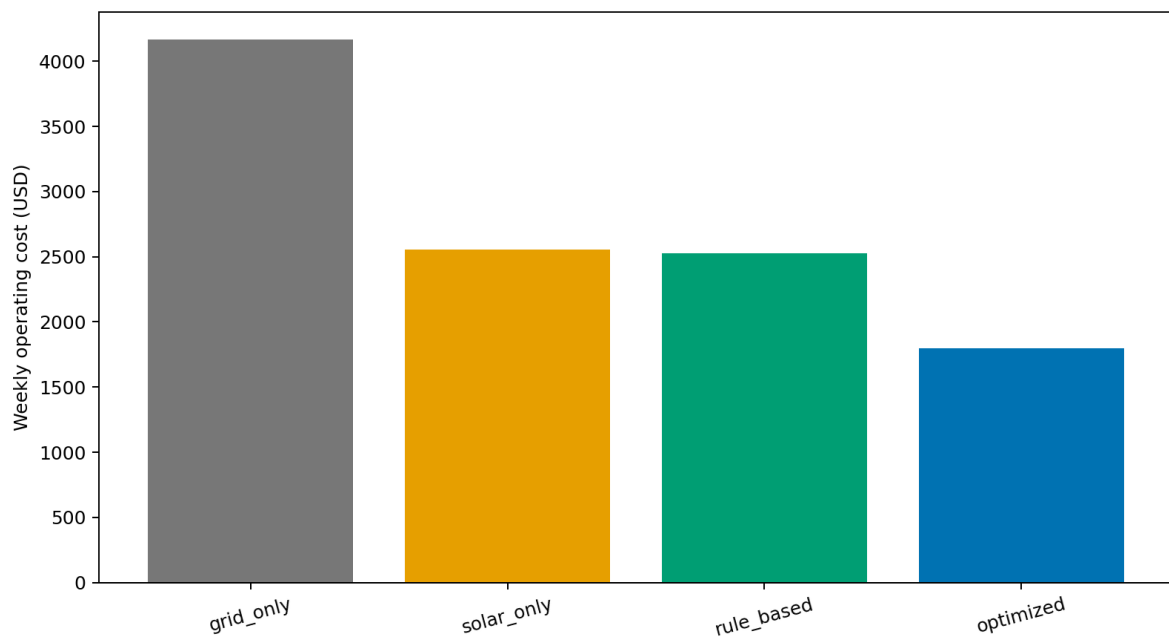
Control and optimization

The rule controller prioritizes PV self-consumption. The MILP minimizes energy cost, demand charge, throughput wear, and unserved-energy penalty subject to equipment, SOC, terminal-energy, grid, and outage constraints. SciPy uses the open-source HiGHS solver.

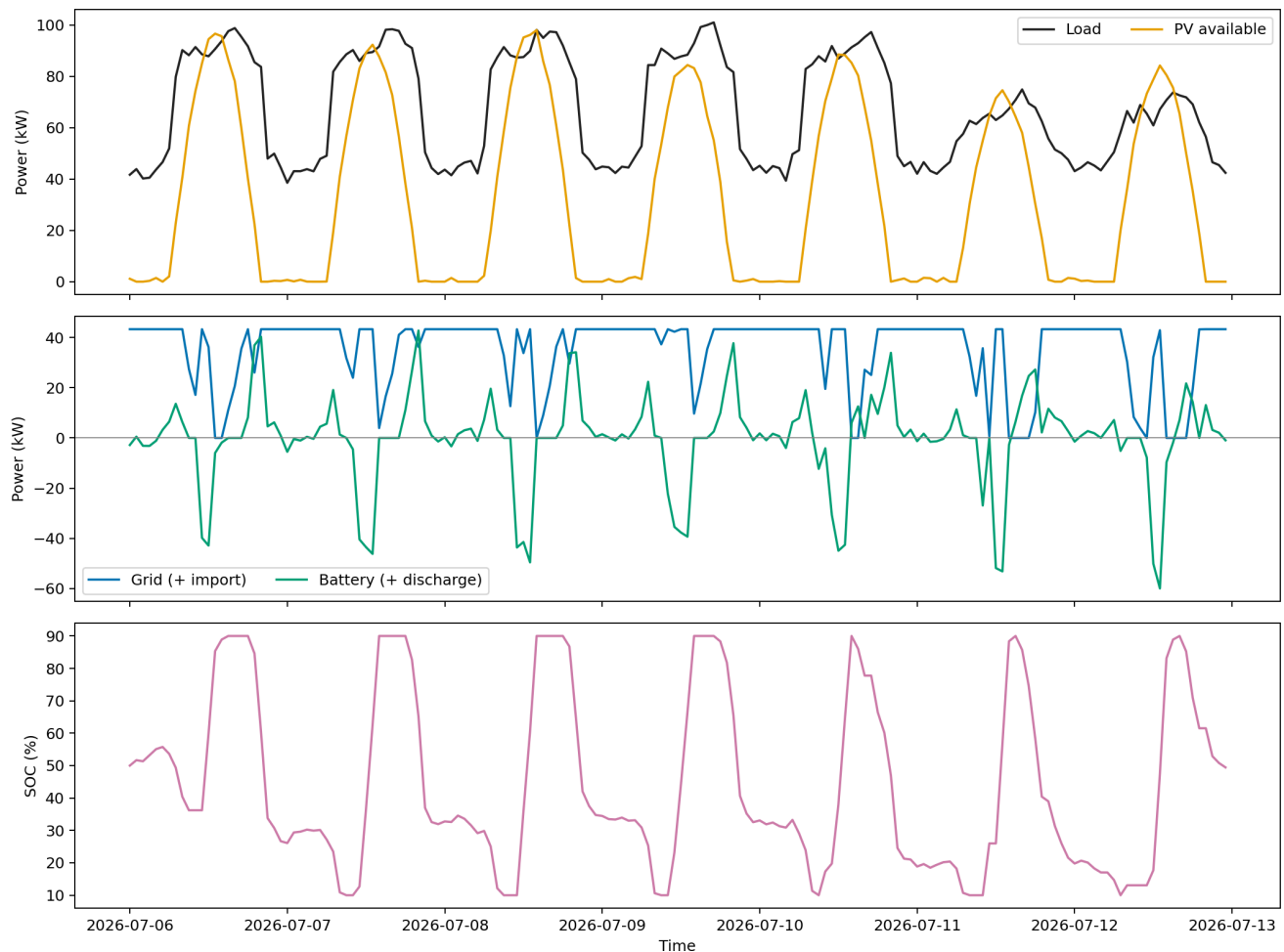
Baseline results

The optimized dispatch achieved the lowest modeled weekly operating cost and reduced peak import. These are simulation results for the supplied synthetic assumptions; they are not a utility bill or field guarantee.

Controller	Import kWh	Peak kW	Total cost	Savings
grid_only	11,317.6	101.0	\$4,170	\$0
solar_only	5,940.2	83.7	\$2,554	\$1,615
rule_based	5,781.2	83.7	\$2,526	\$1,644
optimized	5,922.7	43.4	\$1,800	\$2,370



Optimized dispatch



Validation

All four final controller outputs pass interval-level power balance. The optimized controller passes SOC recursion and limits and has no simultaneous charge/discharge within numerical tolerance. Two automated tests also verify outage isolation.

Limitations and next steps

The model uses perfect forecasts, fixed efficiency, hourly data, and one bus. Future work should add 15-minute measured data, rolling-horizon control, forecast uncertainty, detailed degradation, inverter kVA constraints, and feeder power flow.